

Practical-4

Factorial

Theory :-

I. Iterative Method

The factorial of a non-negative integer n is the product of all positive integers from 1 to n .

$$n! = n \times (n - 1) \times (n - 2) \times \cdots \times 1$$

For example:

$$5! = 5 \times 4 \times 3 \times 2 \times 1 = 120$$

Code :-

```
//Iterative Factorial

#include <iostream>
using namespace std ;

int Factorial(int n ){
    int fact = 1;
    for (int i = 1 ; i <= n ; i++){
        fact *= i ;
    }
    return fact;
}

int main (){
    int n ;
    cout << "Enter a Number : ";
    cin >> n ;

    int result = Factorial(n);

    cout << "Factorial of Number "<< n <<" is "<< result << " .";

}
```

Output :-

```
Problems Output Debug Console Terminal Ports
● PS C:\Users\shiva\Desktop\UNI Coding\DAA> cd "c:\Users
{ .\Practical_4_Iterative }
Enter a Number : 6
Factorial of Number 6 is 720 .
○ PS C:\Users\shiva\Desktop\UNI Coding\DAA> █
```

Analysis Of Algorithm :- Factorial(Iterative)

- Time Complexity: $O(n)$
- Space Complexity: $O(1)$

Theory :-

I. Recursive Method

The recursive factorial method calculates the factorial of a number by calling the same function repeatedly with a smaller value.

The mathematical definition of factorial is:

$$n! = n \times (n - 1)!$$

with the base condition: $1! = 1$

Code :-

```
// Factorial with Recursive method

# include <iostream>
using namespace std;

int Factorial(int n ){
    if (n== 1){
        return 1 ;
    }

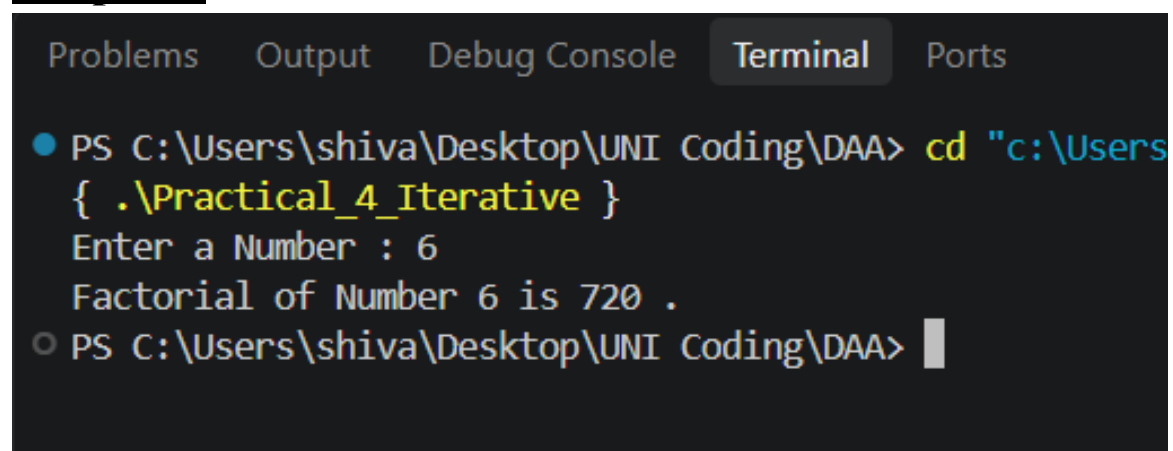
    return n * Factorial(n-1);
}
```

```
int main (){
    int n ;
    cout << "Enter a Number : ";
    cin >> n ;

    int result = Factorial(n);

    cout << "Factorial of Number "<< n <<" is "<< result << " .";
}
```

Output :-



The screenshot shows a C++ IDE with a terminal window. The terminal has tabs for 'Problems', 'Output', 'Debug Console', 'Terminal', and 'Ports'. The 'Terminal' tab is active. The command prompt shows the user navigating to the directory 'C:\Users\shiva\Desktop\UNI Coding\DAA' and running a program. The program prompts for a number, and the user enters '6'. The program outputs 'Factorial of Number 6 is 720 .'.

```
PS C:\Users\shiva\Desktop\UNI Coding\DAA> cd "c:\Users\shiva\Desktop\UNI Coding\DAA"
{ .\Practical_4_Iterative }
Enter a Number : 6
Factorial of Number 6 is 720 .
PS C:\Users\shiva\Desktop\UNI Coding\DAA>
```

Analysis Of Algorithm :- Factorial(Recursive)

- Time Complexity: $O(n)$
- Space Complexity: $O(n)$, due to the recursive call stack